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Homeomorphisms

1.1 Subspaces

Definition 1.1.1

Let Y be a non-empty subset of a topological space $(X, \mathcal{T})$. The collection $\mathcal{T}_Y = \{O \cap Y : O \in \mathcal{T}\}$ of subsets of Y is a topology on Y called the *subspace topology* (or the *relative topology* or the *induced topology* or the *topology induced on Y by $\mathcal{T}$*). The topological space $(Y, \mathcal{T}_Y)$ is said to be a *subspace* of $(X, \mathcal{T})$.